

# Objects

# Exercise

## Storing Values in Objects

Take a look at the code below:

```
# Assign to an object called `x`, the result of `3 + 5`  
x <- 3 + 5  
  
# Assign to an object called `x`, the result of `x - 4`  
x <- x - 4  
  
# Assign to an object called `y`, the result of `x + 1`  
y <- x + 1  
  
# Print the value stored inside of `x`  
print(x)  
  
# Print the value stored inside `y`  
print(y)
```

What are the values of x and y after this code runs?

- a.  $x = 5, y = 5$
- b.  $x = 5, y = 4$
- c.  $x = 4, y = 5$
- d.  $x = 4, y = 4$

# Exercise

## Storing Values in Objects

Take a look at the code below:

```
# Assign to an object called `x`, the result of `3 + 5`  
x <- 3 + 5  
  
# Assign to an object called `x`, the result of `x - 4`  
x <- x - 4  
  
# Assign to an object called `y`, the result of `x + 1`  
y <- x + 1  
  
# Print the value stored inside of `x`  
print(x)  
  
# Print the value stored inside `y`  
print(y)
```

What are the values of x and y after this code runs?

- a. x = 5, y = 5
- b. x = 5, y = 4
- c. x = 4, y = 5
- d. x = 4, y = 4

**Pro Tip:** The lines of code that start with # are called comments. Computers will ignore comments when running your code.

# What are Objects?

**Objects** (also called **variables**) store values for later use.

```
# Assign to an object called `x`, the result of `3 + 5`  
x <- 3 + 5  
  
# Assign to an object called `x`, the result of `x - 4`  
x <- x - 4  
  
# Assign to an object called `y`, the result of `x + 1`  
y <- x + 1  
  
# Print the value stored inside of `x`  
print(x)  
  
# Print the value stored inside `y`  
print(y)
```

- In the exercise, x and y were objects.
- We used the **assignment operator** (<-) to assign values to x and y.

# Rules and Recommendations for Object Names

## Rules

- No spaces (e.g. `num obj`) or mathematical operators (`num-obj`).
- Can't start with underscores or numbers (e.g. `_obj` and `2obj`).
- Can include upper/lower case letters, numbers, underscores, and periods.
- Object names are case sensitive (e.g. `num_obj` and `Num_obj` are not the same).

## Recommendations

- Use **snake\_case**:
  - All lower case letters
  - use underscore(`_`) to separate words
- Use nouns that describe the value that the object contains
  - Avoid using verbs as object names (except for function names).

# Knowledge Check

## Object Names

### Select all that apply:

Which of the following names follow our object name rules?

- a. blood\_pressure
- b. bp
- c. bloodPressure
- d. measure-BP

### Select all that apply:

Which of the following names follow our object name recommendations?

- a. blood\_pressure
- b. bp
- c. bloodPressure
- d. measure-BP

# Knowledge Check

## Object Names

### Select all that apply:

Which of the following names follow our object name rules?

- a. **blood\_pressure**
- b. **bp**
- c. **bloodPressure**
- d. **measure-BP**

### Select all that apply:

Which of the following names follow our object name recommendations?

- a. **blood\_pressure**
- b. **bp**
- c. **bloodPressure**
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